

Ultracold interactions between a single Barium ion and an atomic gas of Lithium

GHE Retreat, Valbella, 2023

Albert-Ludwigs-Universität Freiburg



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What do we do?

- Hybrid traps
- $^{138}\text{Ba}^+$ ions and neutral ^6Li atoms
- Investigate interactions in the ultracold regime

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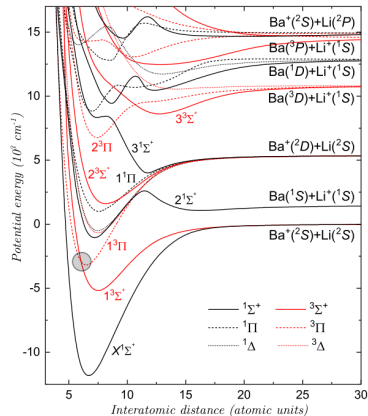
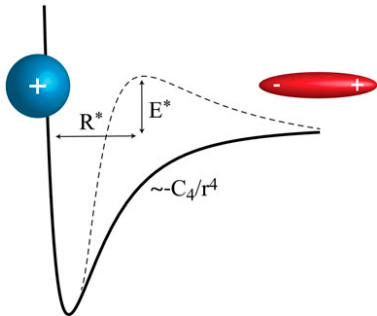
Why hybrid traps?

- Longer range interactions than neutrals (r^{-4} vs r^{-6})
- Cold quantum chemistry
- Quantum simulation

Atom-Ion interaction

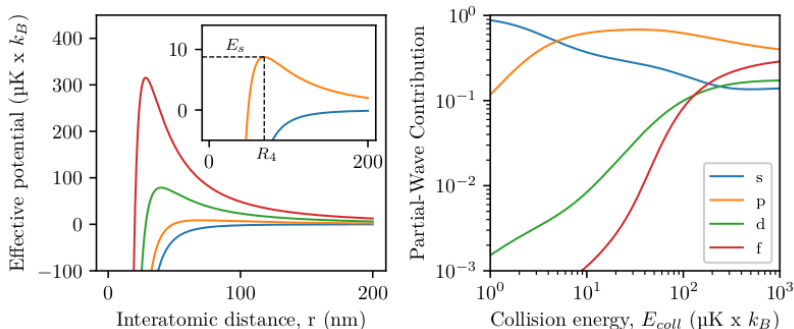
The ion induces a dipole moment in the atom ($p \propto E(r) \propto \frac{1}{r^2}$)

$$V_{\text{atom-ion}}(r) \approx -\frac{C_4}{r^4}$$

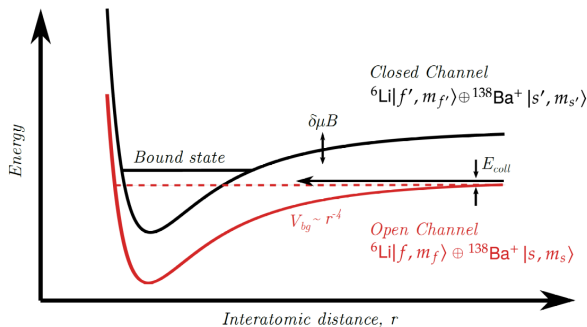


The total wavefunction is composed of partial waves of angular momentum l .
The contribution of partial waves depends on the collision energy.

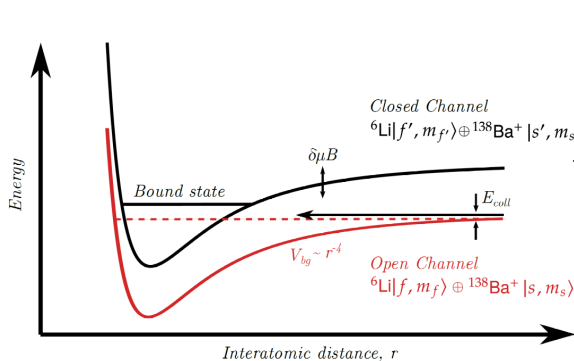
$$V_{\text{atom-ion}}(r) \approx -\frac{C_4}{r^4} + \frac{\hbar l(l+1)}{\mu r^2}$$



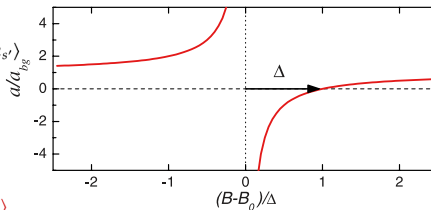
An incoming state (open channel) can couple to a bound state (closed channel).
As the bound state approaches the incoming state the mixing between states increases.



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$$a = a_{bg} \left(1 - \frac{\Delta}{B - B_0} \right)$$



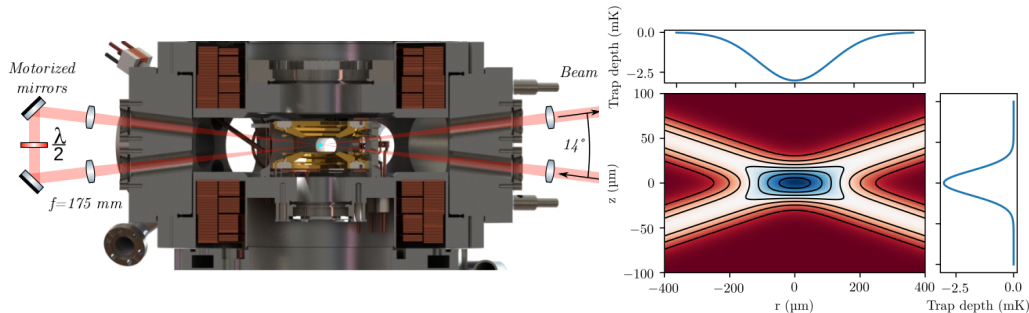
Chin et al. 2010

The Atom Trap

Atoms trapped in a crossed optical dipole trap (xODT).

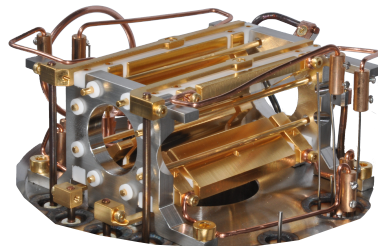
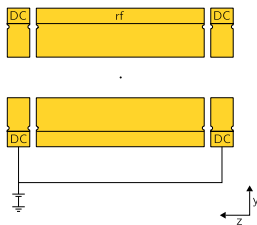
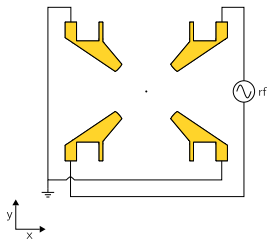
Evaporatively cooled to $T_{\text{Li}} \approx 0.7 - 20 \mu\text{K}$ and spin polarised.

Atom number $N \sim 10^4$, density $n \sim 10^{13} \frac{1}{\text{cm}^3}$.



The Ion Trap

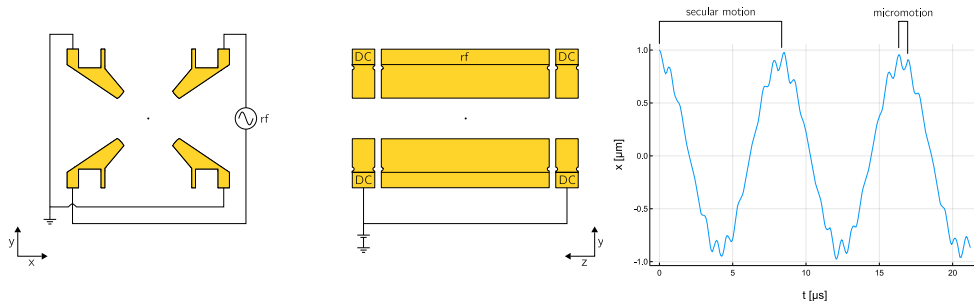
Ion confined in linear rf trap $\Omega_{\text{rf}} = 1.4 \text{ MHz}$, Doppler cooled to $T_{\text{Ba}^+} \approx 300 \mu\text{K}$



Effective potential depth $\sim 10^3 \text{ K}$ traps ion for hours

The Ion Trap

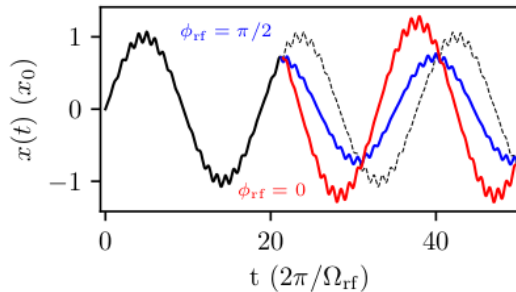
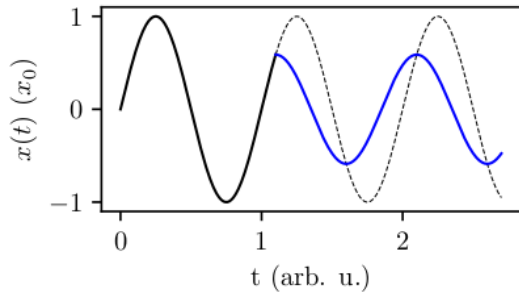
Ion confined in linear rf trap $\Omega_{\text{rf}} = 1.4 \text{ MHz}$, Doppler cooled to $T_{\text{Ba}^+} \approx 300 \mu\text{K}$



Ion's motion includes slower secular part ($\omega_r \approx 100 \text{ kHz}$) and fast microtion part (Ω_{rf})
dc fields increase micromotion:

$$r_i(t) = [1 - q_i \cos(\Omega_{\text{rf}} t)] \cdot [x_{\text{dc}} + x_0 \cos(\omega_r t)]$$

Sympathetic cooling - but ...

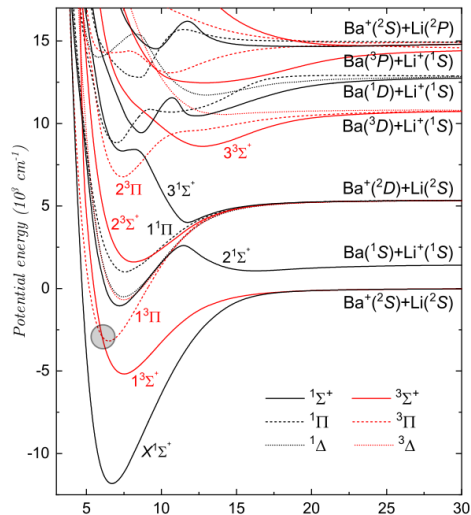


T_{ion} limited by rf trap

- $^{138}\text{Ba}^+$ is hydrogen-like
 - ^6Li is fermionic
 - Large mass ratio
- Collision energy dominated by (colder) Li

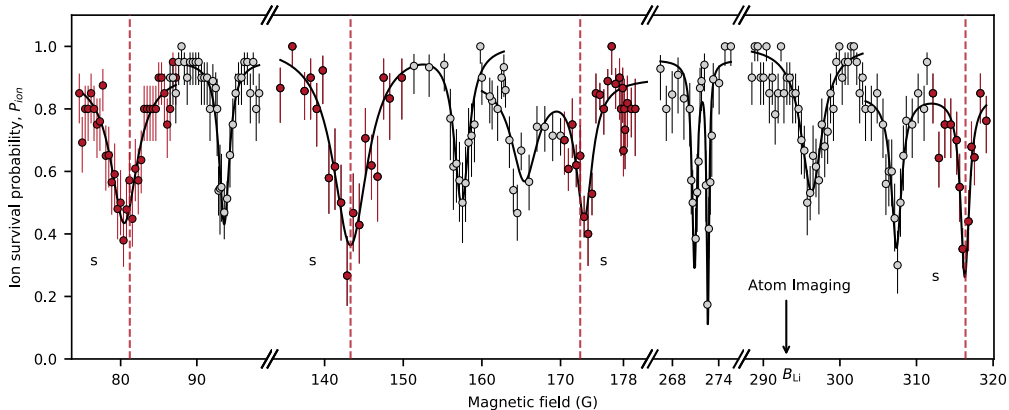
$$E_{\text{coll}} = \frac{\mu}{m_{\text{Li}}} E_{\text{Li}} + \frac{\mu}{m_{\text{Ba}^+}} (E_{\text{Ba}^+} + E_{\text{mm}})$$

- Also reduces the effect collisional heating
- Individual ground states also molecular ground state - low background losses



Feshbach Resonances: Spectrum for Li in $|2\rangle$

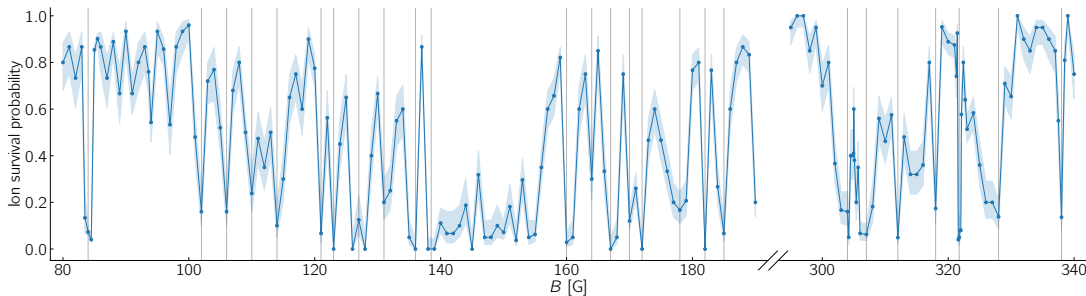
First looked at Li in $|f = 1/2, m_F = -1/2\rangle = |2\rangle$



Free fit of a_S and a_T including coupling to s , p and d assigns four s -wave resonances
(Collaboration with M. Tomza and K. Jachymski in Warsaw)

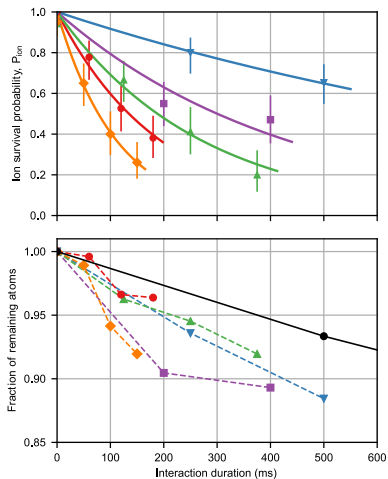
Feshbach Resonances: Spectrum for Li in $|1\rangle$

Then $|f = 1/2, m_F = 1/2\rangle = |1\rangle$



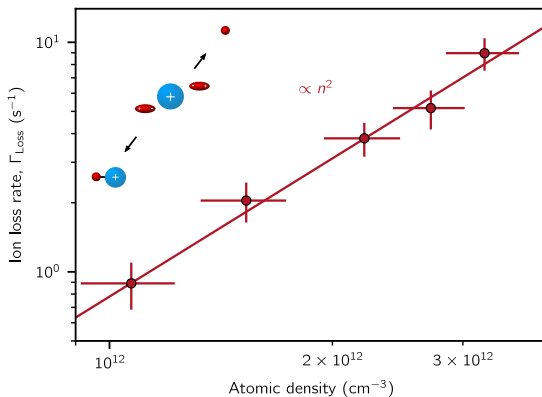
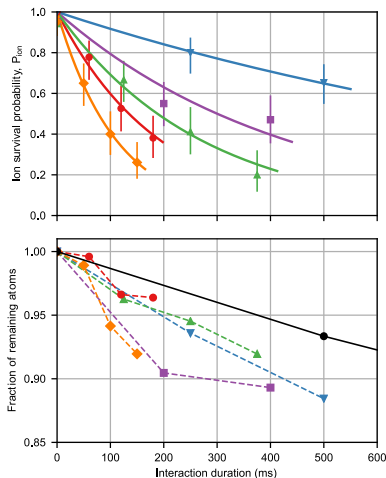
which has a much busier spectrum - more work for the theorists!

Density Dependence of Ba^+ Loss on Resonance



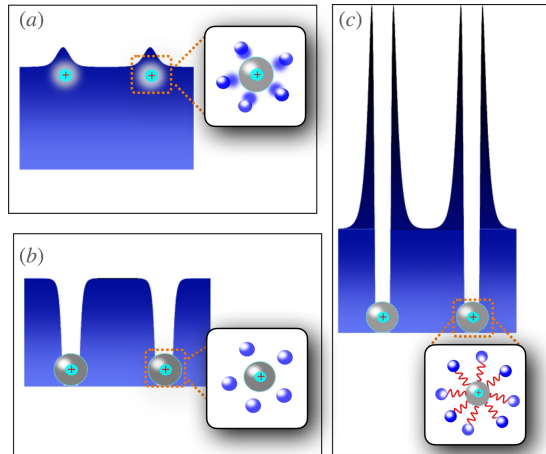
Vary atomic density and probe ion decay rate.

Density Dependence of Ba^+ Loss on Resonance

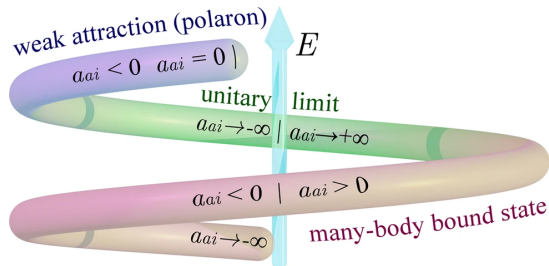


Vary atomic density and probe ion decay rate.
 $\Rightarrow$ TBR is dominant loss process on resonance!

Outlook: Many-Body Physics



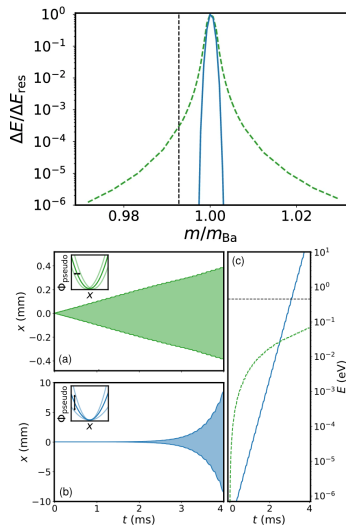
Astrakharchik, Ardila, Jachymski et al.
(2023)



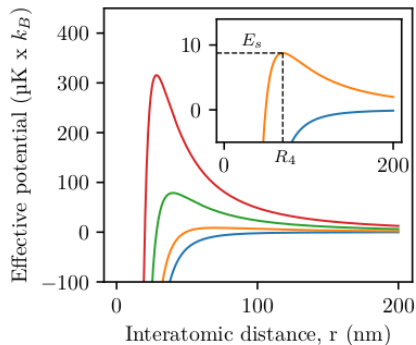
Astrakharchik, Ardila, R. Schmidt et al.
(2021)

- Weak and strong coupling both $\rightarrow \delta m_{\text{eff}}$
 - $\rightarrow$ “tickling” at motional frequency to measure mass
- Weak coupling
 - Able to resolve the shift?
 - $\rightarrow$ Parametric excitation
- Strong coupling
 - Do MBBSs live long enough?

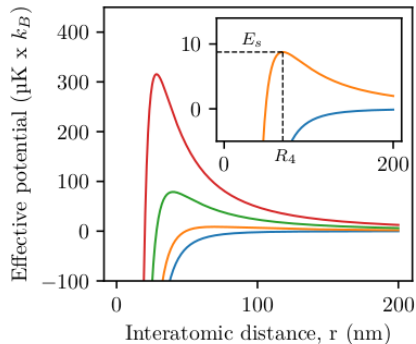
J. Schmidt et al. (2020)



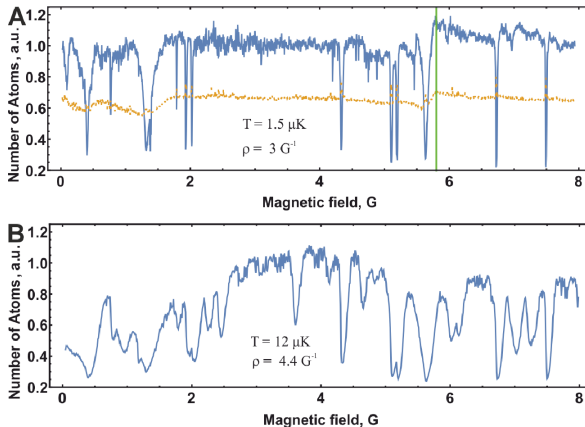
Outlook: Energy Dependence



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Experimental data in thulium:



Khlebnikov et al. (2019)

- Increase ^6Li bath temperature
 - Evaporate to higher temp
 - more light → light assisted interaction?
 - Adiabatic ramps?
- Increase $^{138}\text{Ba}^+$ energy
 - Offset fields to increase micromotion
 - increase in rf power → rf association?
 - displacement → inhomogeneities?
 - Additional rf
 - increase in rf power → rf association?

- Optical trapping of ions demonstrated for $t_{\text{trap}} \gg t_{\text{int}}$
- No micromotion!
- ...but even more light
 - light assisted interactions?

Thank you!



BaLi Team:
Fabian Thieleman, Joachim Welz, Wei
Wu, Pascal Weckesser (former)
Tobias Schätz

Theory collaborators:
Michał Tomza,
Krzysztof Jachymski,
Agata Wojciechowska



Astrakharchik, Grigory E, Luis A Peña Ardila, Krzysztof Jachymski et al. (2023).

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